

Learning Through the Hinge

A student companion for connecting equations across IB Physics

Abstract

Physics equations are compact. A familiar relation can organise several very different situations, while another law, condition or model supplies the physical detail for each one. This guide gives you a method for separating the shared structure from the case-specific physics. The aim is stronger equation choice, clearer explanations and better transfer to unfamiliar problems.

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The central question

What does this equation organise, and what physics completes the case?

Why this helps you

IB questions often change the object, diagram or topic while preserving a deeper mathematical structure. Recognising that structure helps you:

- choose equations from the physics rather than from surface clues;
- connect mechanics, waves, electricity, thermal physics and nuclear physics;
- state the system, direction, controlled variable and assumptions;
- explain where a symbol such as F , P , τ , v or n comes from;
- rebuild a solution when the context is unfamiliar.

A hinge relation is the recurring joint in the argument. It may be a definition, a dynamical law, a conservation statement or a geometrical relation. A second statement supplies the case-specific physics.

$\boxed{\text{shared relation}} + \boxed{\text{case-specific relation}} + \boxed{\text{system and conditions}} \rightarrow \boxed{\text{physical conclusion}}$

A four-step method

1. **Name the shared relation.** Write the equation and define every symbol.
2. **Find the open slot.** Ask which quantity or process remains unspecified.
3. **Supply the case-specific physics.** Add the force law, medium relation, conservation ledger, component law, geometry or state equation.
4. **Declare the conditions.** Identify the system, sign convention, direction, fixed quantity and approximation. Then interpret the result in words.

A quick example: waves

The relation

$$v = f\lambda$$

coordinates wave speed, frequency and wavelength. The medium determines the speed. For a wave on a string,

$$v = \sqrt{\frac{T}{\mu}}.$$

At a boundary, the source frequency remains fixed. A change in medium changes v , so λ changes. The two equations perform different jobs and together give the physical account.

A useful answer frame

The shared relation organises _____. The case-specific relation supplies _____. The result applies when _____.

The hinge map

Hinge	Shared relation	Contexts	Question to ask
Force	$F_{\text{net}} = ma$	gravity; spring; magnetic interaction	Which interaction supplies each force, and which forces belong in the net sum?
Momentum	$F = \Delta p / \Delta t$	collision; rocket; gas pressure	Which system is being followed, and how does momentum enter, leave or change?
Torque	$\tau_{\text{net}} = I\alpha$	gravity; friction; magnetic interaction	Which force has a moment about the chosen axis, and what is its perpendicular lever arm?
Power	$P = \Delta E / \Delta t$	lifting; resistor; lamp	Which energy pathway is being measured, and which relation gives the transfer rate?
Waves	$v = f\lambda$	string; sound; light in a medium	Which property of the medium determines v , and which quantity stays fixed at a boundary?
Energy	signed energy ledger	circuit loop; thermal process; nuclear binding	Which transfers and stores belong in the ledger, and what gives each term its value?
Flux	total and effective area	pressure; solar irradiance; magnetic flux	What is distributed across the surface, and how does orientation change the effective area?
Number density	number divided by space	grating lines; charge carriers; ideal gas	How does the count per length or volume enter the physical law for the observable?

What strong solutions contain

A complete solution usually includes four layers.

Shared structure	The governing relation, correctly written and interpreted.
Case-specific physics	The interaction law, material relation, geometry, conservation rule or state equation that completes the model.
Conditions	The chosen system, direction, sign convention, fixed quantities and relevant approximation.
Physical meaning	A sentence explaining the calculated result and its dependence on the variables.

Three checks before finishing

- **Units:** do both sides and the final value have the required dimensions?
- **Signs and directions:** does the sign match the chosen convention and physical process?
- **Dependence:** can you explain what happens when one variable increases while the stated conditions are held fixed?

Equation status matters

State the role of the relation you are using. Examples include a definition, conservation condition, dynamical law, interaction law, constitutive relation, geometrical relation or approximation. This prevents algebraically similar equations from being treated as interchangeable explanations.

Using the worksheet set

For each worksheet, complete the three cases and then write one synthesis paragraph. Your paragraph should identify:

- the structure that remains the same;
- the physical relation that changes from case to case;
- the system or condition controlling the interpretation;
- one new context in which the same hinge could be useful.

Energy	hinge_energyHinge.pdf
Flux	hinge_fluxHinge.pdf
Force	hinge_forceHinge.pdf
Momentum	hinge_momentumHinge.pdf
Number density	hinge_numberDensity.pdf
Power	hinge_powerHinge.pdf
Torque	hinge_torqueHinge.pdf
Waves	hinge_waveHinge.pdf

The learner payoff

A formula sheet gives you relations. Physics understanding lets you assemble them into a model. The hinge approach trains the assembly process: recognise the invariant structure, supply the local physics, state the conditions and interpret the result.

When facing an unfamiliar problem, ask:

Which part of this problem is structurally familiar, and what must I add to make the model specific?